

## 7. Phase B characterization slides — plain-English walkthrough

Last synced: 2026-08-04 (5heat / 5line split)

Audience: Anyone who has **not** been in the weeds — advisor, collaborator, or future-you after time away.

Deck: [HEROs\\_and\\_PASSes/slides/CHAR\\_Phase\\_B\\_characterization.pptx](#) | (hand-edited master).

Slide count: Auto merge = **7** slides. Charles’s hand deck may be **8** —  $\theta \times K/N$  is split into **5heat** (heatmap) and **5line** (line plot); delete one after review if desired. Slides **6–7** below =  $\gamma$  sweep +  $\lambda_{\text{crit}}$  (renumber to **7–8** if both  $\theta \times K/N$  slides stay in).

What this document is: A slide-by-slide explanation of **what you are looking at** and **why it matters** for the dissertation story. It does not replace the technical memos; it orients a reader before they open PowerPoint.

Companion: [CHARLES\\_CHECKLIST.md](#) (what Charles still owes); [06\\_Lambda\\_threshold\\_and\\_KN\\_memo.md](#) (sort-and-chop  $\lambda$  threshold detail).

### Intro — what this deck is trying to do

#### The big picture (one paragraph)

Real college basketball data show a striking pattern: players on **stronger teammate pools** do not always have the **highest** NBA draft rates — the relationship can **bend** (rise, flatten, even dip at the very top). Charles’s dissertation asks whether a **simple fake league** can reproduce shapes like that when we write down explicit rules: put players on teams, **score** them, **select** the top few, then **look at who got selected** broken down by roster context.

**Phase B** (this deck) is **not** “fit the real hero yet.” It is **characterization**: turn one dial at a time in the fake league and ask, *does this dial matter for the shape of the selection curve?* If yes, the dial is part of the story; if no, we note that and move on.

#### Three steps in the fake league (memorize this)

Every generative slide shares the same pipeline:

1. **ASSIGN** — Who plays where? (team matching / sorting; knob  $\rho$ )
2. **SCORE** — Who looks best on paper?  $S_i = A_i - \lambda \cdot L_C$  (knob  $\lambda$ ; congestion term  $L_C$ )
3. **SELECT** — Who actually advances? Top  $K$  players by score (knob  $K/N$  = selectivity)

Then we **VISUALIZE**: among the  $K$  selected players, plot the **fraction selected in each bin** of **team pool quality** (x-axis = mean ability on the roster, including self — called **pool mean** here).

So the y-axis is **not** “draft rate in the real NCAA.” It is “**share of selected players in this bin**” in a synthetic league with  $N \approx 5,600$  players and default  $K/N = 10\%$  (~560 selected), unless a slide says otherwise.

#### How to read the curves

- **X-axis (pool mean bins)**: Teams sorted into 16 equal-sized bins from **weakest** average talent (left) to **strongest** (right). Bin 1 = low pool mean; bin 16 = high pool mean.
- **Y-axis**: For each bin, what **fraction of all selected players** landed in teams whose pool mean falls in that bin? High values mean “selection concentrates in this roster environment.”
- **Monotone rising curve**: Elite team environments hog selection — looks like “the best teams produce the picks.”
- **Hump / inverted-U-ish curve**: Selection peaks in **middle** pool-mean bins — “mid-tier pressure” shows up in who gets through.
- **Flat curve**: Assignment or score barely changes **where** selected players sit on the pool-mean ladder.

#### Knobs on the slides (mini glossary)

| Symbol    | Plain name                        | What it controls                                                        |
|-----------|-----------------------------------|-------------------------------------------------------------------------|
| $A_i$     | Player ability                    | Innate talent draw (here: Beta(2,2) on [0,1])                           |
| $\rho$    | Assignment assortativity          | How tightly players are matched to team talent targets                  |
| $\lambda$ | Congestion weight in <b>score</b> | How much peer viability $L_C$ penalizes ranking                         |
| $L_C$     | LOO peer viability congestion     | Mean over teammates $j \neq i$ of $\sigma(\gamma(A_j - \theta))$        |
| $\theta$  | Viability cutline                 | Center of the sigmoid — who counts as a “viable” peer                   |
| $\gamma$  | Sigmoid sharpness                 | How sharp the viability step is around $\theta$                         |
| $K/N$     | Selectivity                       | Fraction of the league that gets selected (slots $K \div$ players $N$ ) |

**One-at-a-time (OAT)**: Each slide fixes **everything except one knob** (or one small grid for  $\theta \times K/N$ ). Same random seed (42) where possible so differences are from the knob, not a new lottery.

### SLIDE 1 — Characterize $\rho$ (assignment): sort-and-chop shown

**Knob varied**:  $\rho$  — how assortative assignment is (plus a **sort-and-chop** benchmark arm).

**Held fixed**: Score rule  $S_i = A_i - 0.55 \cdot L_C$ , top- $K$  selection, same draw of abilities and team targets.

#### What the figure shows

Several colored curves on one plot. Each curve is a different **assignment rule** applied to the **same** pool of player talent:

- **Low  $\rho$  (e.g. 0.1)**: Players mix almost randomly across team talent levels → selected players spread evenly across pool-mean bins → curve stays \*\* fairly flat\*\*.

- **Moderate / high  $\rho$  (1, 8, 32)**: Strong players tend to land on strong teams  $\rightarrow$  selection shifts toward high pool-mean bins  $\rightarrow$  curve can develop a **peak** or strong tilt to the right.
- **Sort-and-chop arm**: A **hard** sorting benchmark (perfect rank matching within slices). Included here for comparison — it often produces an extreme **monotone** shape (selection piles into the top bins).

**What we learn**

**Who you play with (assignment) matters a lot** once score and selection are fixed. Low mixing  $\rightarrow$  flat readout; high assortativity  $\rightarrow$  peer environments concentrate and the selection-vs-pool-mean curve **bends**.

**What we are *not* claiming**

This is a **qualitative** proof-of-concept in a fake league. The x-axis is **pool mean**, not the empirical hero axis **poolq\_loo**. We have not yet matched bin-for-bin to real NCAA draft rates.

**SLIDE 2 — Characterize  $\rho$  (assignment): sort-and-chop suppressed**

Same experiment as Slide 1, but the **sort-and-chop** curve is **omitted from the plot** so the eye focuses on the **soft  $\rho$  ladder** only ( $\rho \in \{0.1, 1, 8, 32\}$ ).

**Why a second slide**

Sort-and-chop is a useful **benchmark** (zero overlap between talent slices), but it can **dominate** the figure visually. Soft  $\rho$  arms are closer to “continuous” assortativity and are easier to compare to each other.

**What we learn**

Same claim as Slide 1, stated without the hard-sort distraction: **raising  $\rho$**  moves the selection readout from **flat** toward **concentrated / humped** shapes on the pool-mean axis.

**SLIDE 3 — Characterize  $\lambda$  (congestion in score)**

**Knob varied**:  $\lambda$  in  $S_i = A_i - \lambda \cdot L_C$ .  
**Held fixed**: Soft assignment at  $\rho = 8$ , same rosters and talent draw, same top-K rule.

**Arms on the plot**

Typically four values:

| $\lambda$ | Score rule       | Plain English                                      |
|-----------|------------------|----------------------------------------------------|
| 0         | $S_i = A_i$      | <b>Talent only</b> — congestion ignored in ranking |
| 0.25      | mild penalty     | Light peer pressure in score                       |
| 0.55      | moderate penalty | Default-style weight in many Pass B/C runs         |
| 1.0       | strong penalty   | Heavy peer viability penalty                       |

**What the figure shows**

- **$\lambda = 0$** : Curve tends **monotone** — players on the highest pool-mean teams dominate selection because score  $\approx$  raw ability and strong players sit on strong teams.
- **$\lambda > 0$** : Congestion enters the **ranking**. Players in extremely strong peer fields can be **penalized** in score even if they are good  $\rightarrow$  selection can **spread** or **peak** away from the very top bins  $\rightarrow$  curve **bends** (inverted-U-ish readout on pool mean).

**What we learn**

**Congestion belongs in the score, not just in the environment story**. If advancement used talent alone ( $\lambda = 0$ ), you would not get the same selection shape as when roster pressure enters  $S_i$ .

**Connection to real data (conceptual only)**

The empirical hero bins on **teammate quality** in **outcomes**. This slide bins on **pool mean** in a **simulated selection** step. Same *kind* of question (“does context bend who gets through?”), different measurement and league.

**SLIDE 4 — Characterize  $\theta$  (viability cutline): one-at-a-time at  $K/N = 10\%$**

**Knob varied**:  $\theta$  — center of the logistic viability function inside  $L_C$ .  
**Held fixed**:  $\rho = 8$ ,  $\lambda = 0.55$ ,  $\gamma = 10$ ,  **$K/N = 10\%$**  (characterization default).

**The equation on the slide (what  $\theta$  does)**

$$L_C = \text{mean}_{j \neq i} \sigma(\gamma(A_j - \theta))$$

For each player  $i$ , look at teammates  $j$ , map each teammate’s ability through a **soft step**  $\sigma$  centered at  $\theta$ , average (leave-one-out). **Higher  $\theta$**   $\rightarrow$  fewer teammates count as “viable peers”  $\rightarrow$   $L_C$  changes  $\rightarrow$   $S_i$  changes  $\rightarrow$  selection shifts.

**Arms**

$\theta \in \{0.50, 0.72, 0.90\}$  — low, default-ish (0.72), high cutline.

What the figure shows (typical pattern)

- $\theta = 0.50$ : More peers count as viable  $\rightarrow$  congestion term behaves differently  $\rightarrow$  selection peak often sits in **mid** pool-mean bins (e.g. peak around bin 13).
- $\theta = 0.72$  (539 default): Similar hump, sometimes slightly higher peak rate.
- $\theta = 0.90$ : Viability is hard to achieve  $\rightarrow$  congestion term shifts  $\rightarrow$  selection can **pile into top bins** (peak at bin 16) — curve looks more **monotone** at the elite edge.

What we learn

$\theta$  is not a cosmetic parameter. Moving the viability cutline changes **who wins** under fixed  $\rho$ ,  $\lambda$ , and selectivity. Before tying  $\theta$  to real draft data, we need to see it move the sim curve — it does.

SLIDE 5heat — Characterize  $\theta \times K/N$  (heatmap)

Hand deck: usually slide 5. PNG: [theta/THETA\\_KN\\_sweep\\_peak\\_bin.png](#)

Knobs varied together (small grid):  $\theta \times K/N$  — on purpose, because PD15 asked whether viability cutline and selectivity **co-move**.

Held fixed:  $\rho = 8$ ,  $\lambda = 0.55$ ,  $\gamma = 10$ ,  $N = 5600$ ; only **K** changes with  $K/N$ .

What this figure is

A **3×3 heatmap** of **peak pool-mean bin** (integer 1–16). We do **not** plot full selection curves here — only *where the hump peaks* on the pool-mean ladder for each ( $\theta$ ,  $K/N$ ) cell.

- **Rows:**  $K/N$  presets — MBB-like **1%**, characterization **10%**, high **40%**.
- **Columns:**  $\theta \in \{0.50, 0.72, 0.90\}$ .
- **Cell color / number:** Peak bin (1 = weakest team environments, 16 = strongest).

Example readout (seed 42)

| K/N regime              | $\theta = 0.50$ | $\theta = 0.72$ | $\theta = 0.90$ |
|-------------------------|-----------------|-----------------|-----------------|
| ~1% (MBB-like)          | 6               | 9               | 12              |
| ~10% (default)          | 13              | 13              | 16              |
| ~40% (high selectivity) | 16              | 16              | 16              |

How to read the heatmap

- **Yellow / high numbers ( $\rightarrow 16$ ):** Selection peak sits on **elite** team environments — top-saturated.
- **Green / mid numbers (~6–13):** Peak sits **mid-ladder** — inverted-U-ish readout on pool mean.
- **Scan a row (fixed  $K/N$ ):** See whether  $\theta$  moves the peak as you go left  $\rightarrow$  right.
- **Scan a column (fixed  $\theta$ ):** See whether selectivity moves the peak as you go top  $\rightarrow$  bottom.

When to use this slide

Best when Alex (or you) want **exact bin numbers** at a glance, or when comparing to the table in [theta/THETA\\_KN\\_sweep\\_summary.csv](#) without estimating from a line chart.

What we learn (shared with 5line)

**Do not pick  $\theta$  in isolation.** At 1% selectivity, raising  $\theta$  shifts the peak up the pool ladder (6  $\rightarrow$  9  $\rightarrow$  12). At **40%**, everything saturates at bin **16** — selectivity **swamps**  $\theta$ . At **10%**,  $\theta$  toggles mid-hump vs top-saturation (13 vs 16). **Descriptive only** — not a fitted law  $\theta = f(K/N)$ .

SLIDE 5line — Characterize  $\theta \times K/N$  (line plot)

Hand deck: duplicate of 5heat, usually slide 6. PNG: [theta/THETA\\_KN\\_sweep\\_peak\\_bin\\_lines.png](#)

Same experiment as 5heat — identical 3×3 grid, different graphic.

What this figure is

Peak bin vs  $\theta$ , with **one line per  $K/N$  preset**:

- **Green — MBB-like ( $K/N \approx 1\%$ ):** Peak rises with  $\theta$  ( $\approx 6 \rightarrow 9 \rightarrow 12$ ) —  $\theta$  and selectivity **interact** when few slots exist.
- **Blue — characterization ( $K/N \approx 10\%$ ):** Mostly flat then jumps to bin 16 at high  $\theta$  (13, 13, 16).
- **Red — high selectivity ( $K/N \approx 40\%$ ):** Flat at bin **16** for all  $\theta$  — everyone elite is already selected.

Markers sit at the **three measured  $\theta$**  values only; line segments connect those points. **Not** a dense surface or interpolated grid — only the nine diagnostic runs.

How to read the line plot

- **Steep green line:**  $\theta$  matters a lot when the league is highly selective (few picks).
- **Flat red line at top:** When many advance,  $\theta$  barely moves where the peak sits.
- **Subtitle on figure:** “3×3 diagnostic grid — lines join measured  $\theta$  only” — remind audience we are not claiming smooth  $\theta$ – $K/N$  physics between grid points.

When to use this slide

Best for **live presentation** — the co-variation story is visible in one glance. Use **instead of 5heat** in a tight meeting, or keep both and hide one in PowerPoint until you decide.

What we learn

Same as **5heat**; the claim in your footer applies to both:

*$\theta$  and  $K/N$  co-vary — at MBB-like selectivity, raising  $\theta$  shifts peak bin up the pool ladder; at 40% selectivity, curves are top-saturated regardless of  $\theta$ .*

SLIDE 6 — Characterize  $\gamma$  (viability sharpness): selection curves

**Knob varied:**  $\gamma$  — steepness of  $\sigma(\gamma(A - \theta))$  inside **L\_C**.  
**Setting:** **Sort-and-chop** assignment benchmark (same rosters within each  $\gamma$ ); several  $\lambda$  arms overlaid per  $\gamma$ .

Arms

- $\gamma \in \{5, 10, 20\}$  — soft, default, sharp viability step
- For each  $\gamma$ ,  $\lambda \in \{0, 0.25, 0.55, 0.75, 1.0\}$

What the figure shows

Each panel or color group asks: *as we turn up congestion weight  $\lambda$ , when does the selection curve stop looking like  $\lambda = 0$ ?*

- Low  $\gamma$  (5):** Viability spreads over ability  $\rightarrow$  congestion builds slowly  $\rightarrow$  need **larger  $\lambda$**  before curves change shape.
- High  $\gamma$  (20):** Sharp knee  $\rightarrow$  congestion “turns on” quickly  $\rightarrow$  **smaller  $\lambda$**  can reorder selection.

What we learn

$\gamma$  sets how “binary” peer viability feels. Sharper viability  $\rightarrow$  congestion bites at lower  $\lambda$ . This feeds directly into Slide 7’s  $\lambda_{crit} \approx 4/\gamma$  rule of thumb on sort-and-chop (Slide 8 if both  $\theta \times K/N$  variants remain in the hand deck).

SLIDE 7 —  $\lambda_{crit} \approx 4/\gamma$  (educational)

**Not a new sweep** — an **explanatory** figure tying  $\gamma$  to a **critical  $\lambda$**  where score ranking first differs from talent-only ranking.

What the figure shows

- The **sigmoid**  $\sigma(\gamma(A - \theta))$  along the ability axis for several  $\gamma$ .
- Vertical markers at  $\lambda_{crit} \approx 4/\gamma$  (e.g.  $\gamma = 10 \rightarrow crit \approx 0.4$ ).

Plain-English meaning of  $\lambda_{crit}$

On sort-and-chop rosters, ability and congestion **line up** within teams. Small  $\lambda$  may **not reorder** who makes top-**K** even though  $\lambda > 0$  in the formula.  $\lambda_{crit}$  is the first  $\lambda$  where  **$S_i = A_i - \lambda L_C$**  actually changes **who** gets selected vs  **$S_i = A_i$** .  
**Important nuance:** The **plot** in 16 bins may still **look** like  $\lambda = 0$  until  $\lambda$  is **slightly above**  $\lambda_{crit}$  — coarse bins hide small reorderings. Finer bins or checking rank swaps tell the fuller story (see memo 06).

Why this slide exists

Alex asked why  $\lambda = 0$  and  $\lambda = 0.25$  can look **identical** on some diagnostics. This slide answers: *because 0.25 is below the reorder threshold for  $\gamma = 10$  on sort-and-chop*, not because  $\lambda$  “does nothing” in the math.

How the slides fit together

| Slide | Knob                | One-line takeaway                                                                       |
|-------|---------------------|-----------------------------------------------------------------------------------------|
| 1–2   | $\rho$              | Sorting / assignment shapes roster environments $\rightarrow$ changes selection readout |
| 3     | $\lambda$           | Congestion in <b>score</b> bends the curve vs talent-only                               |
| 4     | $\theta$            | Viability cutline moves the hump at fixed $K/N = 10\%$                                  |
| 5heat | $\theta \times K/N$ | Heatmap — exact peak bins; $\theta$ interacts with selectivity                          |
| 5line | $\theta \times K/N$ | Same grid as 5heat — trend lines for presenting                                         |
| 6     | $\gamma$            | Sharpness of viability sets when $\lambda$ starts to matter                             |
| 7     | $\lambda_{crit}$    | Explains “identical” low- $\lambda$ curves on sort-and-chop                             |

*If both 5heat and 5line stay in the hand deck,  $\gamma$  and  $\lambda_{crit}$  become slides 7 and 8.*

**Story arc for an outsider:** Real data show context matters in **outcomes**. This deck shows which **rules of a fake league** are powerful enough to bend **who gets selected** — assignment ( $\rho$ ), congestion in score ( $\lambda$ ), viability geometry ( $\theta$ ,  $\gamma$ ), and league selectivity ( $K/N$ ). Phase C (later) asks whether we can **fit** those knobs to data; Phase B asks whether the **model has the right moving parts**.

Practical notes for the reader

- Hand vs auto decks:** Scripts regenerate disposable `slides/auto/*_AUTO.pptx`; Charles’s formatted master is `slides/CHAR_Phase_B_characterization.pptx`. See `slides/README.txt` for PNG paths if figures are refreshed.
- Regenerate figures only:** `./scripts/build_characterization_slides.sh` (does not touch the hand deck).
- Empirical hero lives elsewhere:** Pass A / `So_Far_.pptx` slides 1–3 — real MBB draft rates vs talent and poolq\_loo.

Suggested one-sentence opener if you present this deck

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"We built a transparent fake league, varied one rule at a time, and asked which rules bend **who gets selected** when we bin by team strength — assignment  $\rho$ , congestion  $\lambda$  in the score, viability  $\theta$  and  $\gamma$ , and selectivity  $K/N$  all matter in ways this deck summarizes."